ORGANIC CHEMISTRY-II

UNIT 4 NOTES

POLYNUCLEAR HYDROCARBONS

- NAPTHALENE
- PHENANTHRENE
- ANTHRACENE
- DIPHENYLMETHANE
- TRIPHENYLMETHANE

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POLYNUCLEAR HYDROCARBONS

- Polynuclear Hydrocarbons are belongs to the class of Aromatic Hydrocarbons that contains multiple Aromatic rings made up of Hydrocarbon compounds.
- Two or more benzene rings fused together to form these compounds.
- These compounds are also known as Polycyclic Aromatic Hydrocarbons.

TYPES

Polynuclear Hydrocarbons are mainly divided into two types :

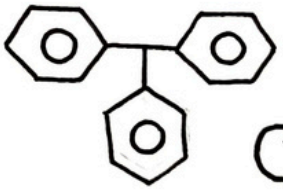
① Isolated Ring Compounds
② Fused Ring Compounds

① ISOLATED RING COMPOUNDS

In these compounds Aromatic rings are isolated.



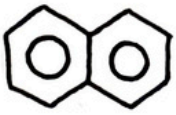
DIPHENYL METHANE



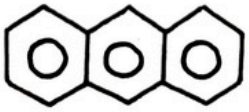
TRIPHENYL METHANE

② FUSED RING COMPOUNDS

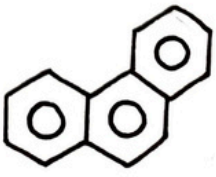
In these compounds Aromatic Rings are Fused.



NAPTHALENE



ANTHRACENE



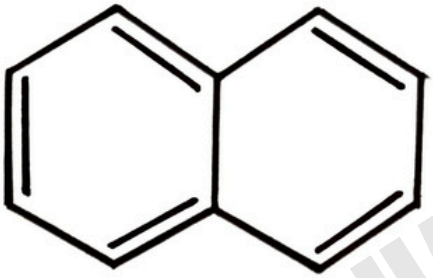
PHENANTHRENE

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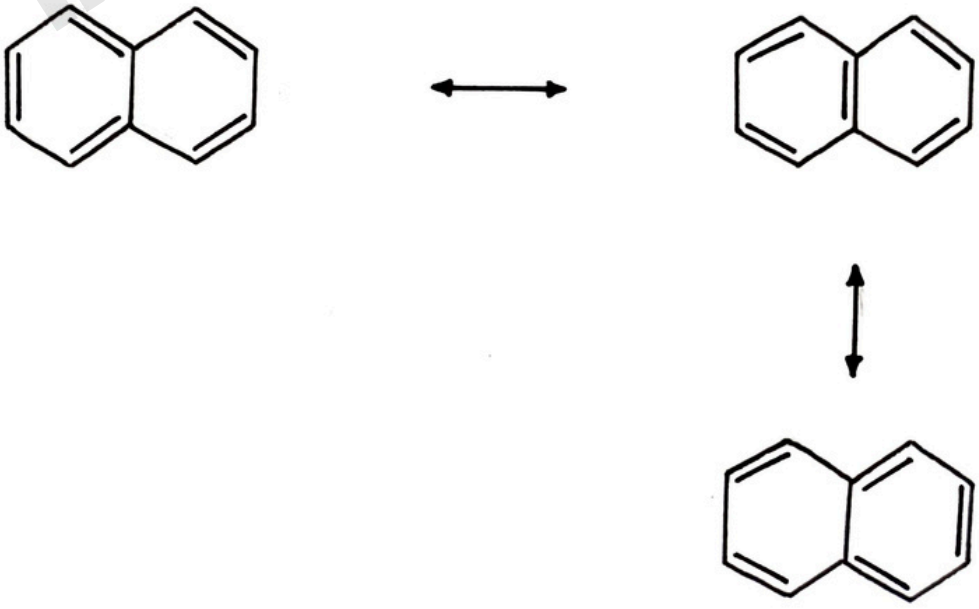
NAPTHALENE

- A Napthalene molecule is derived from fusion of pair of Benzene Ring.
- It is a member of fused ring hydrocarbons under the class of Polynuclear Hydrocarbons.
- Napthalene is a crystalline , aromatic , white solid hydrocarbon.



RESONANCE IN NAPTHALENE

- Napthalene forms 3 resonating structure.
- Delocalization of π bonds makes napthalene stable.

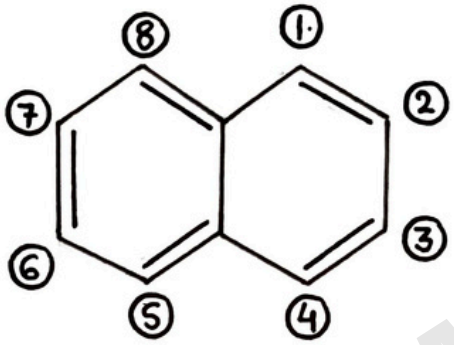


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CHEMICAL PROPERTIES

- Chemical Formula of Napthalene is $C_{10}H_8$
- All carbons in napthalene are sp^2 hybridized.
- Napthalene are more reactive than benzene because resonance energy in napthalene (61 kcal/mol) is less than twice to that of benzene (36 kcal/mol).



- In the above structure of Napthalene Position 1, 4, 5, 8 → α
  and Position 2, 3, 6, 7 → β
- Also Unlike the Benzene, C-C bond length in napthalene are not same.
- C_1-C_2 bond length in napthalene is shorter (1.36 Å) than C_2-C_3 bond length (1.40 Å) due to more double bond character on C_1-C_2 bond.

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PHYSICAL PROPERTIES

- Napthalene is a white crystalline solid.
- Its Melting Point is 80°C.
- Its Boiling Point is 217°C.
- It is insoluble in water but soluble in organic solvents.

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SYNTHESIS

Napthalene can be synthesized by following various methods :

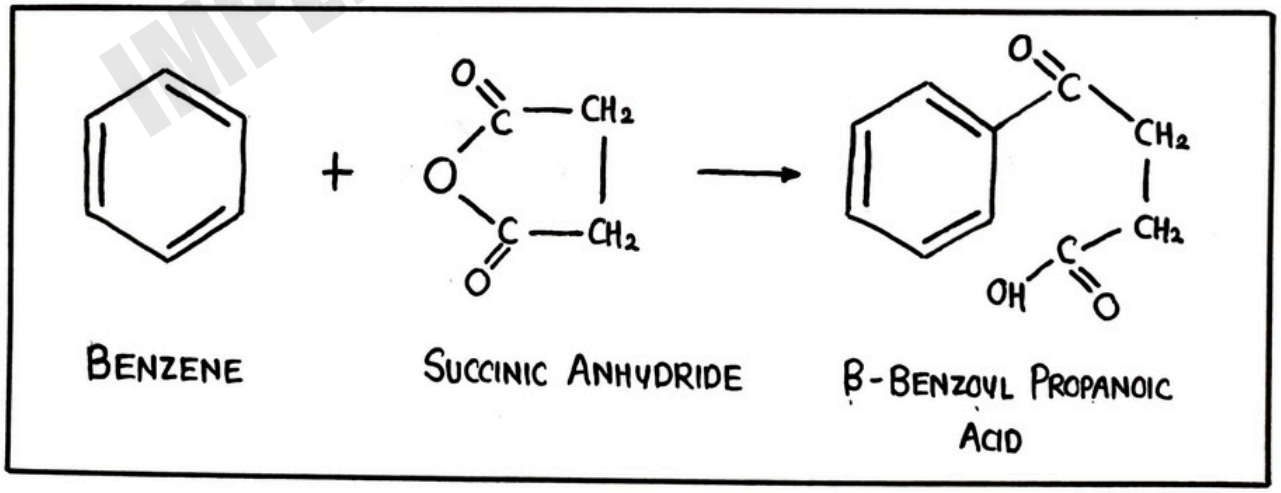
① HAWORTH SYNTHESIS

It is a Five step chemical reaction :

- Friede - Craft Acylation
- Clemmensen Reduction - I
- Ring Closure Reaction
- Clemmensen Reduction - II
- Aromatization

- **STEP - I (Friedel Craft Acylation)**

Benzene and succinic Anhydride are heated in the presence of $AlCl_3$ to form 3- benzoyl Propanoic Acid.

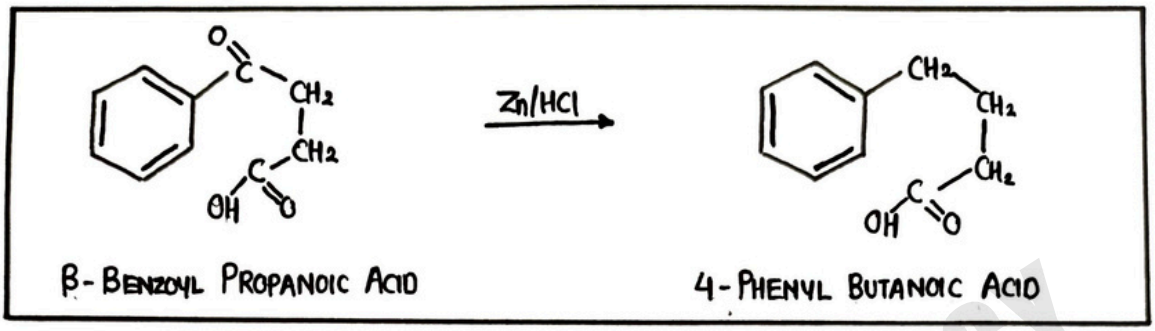


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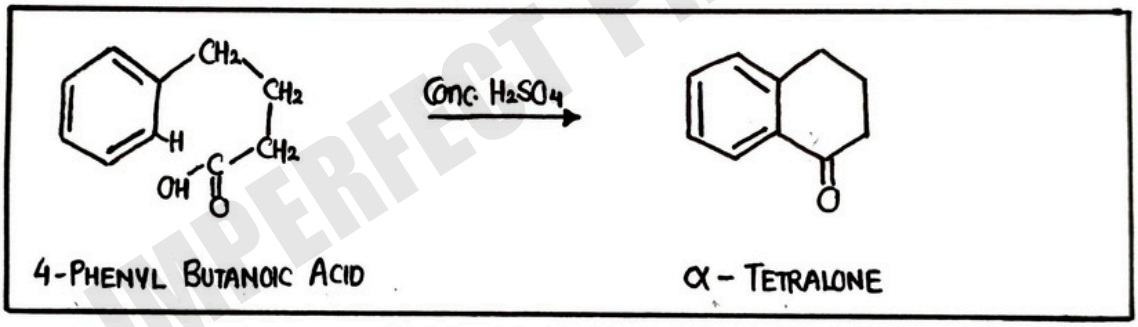
- **STEP - II (Clemmensen Reduction)**

β- Benzoyl Propanoic Acid is treated with Amalgamated zinc in the presence of HCl to give 4- Phenyl Butanoic Acid.



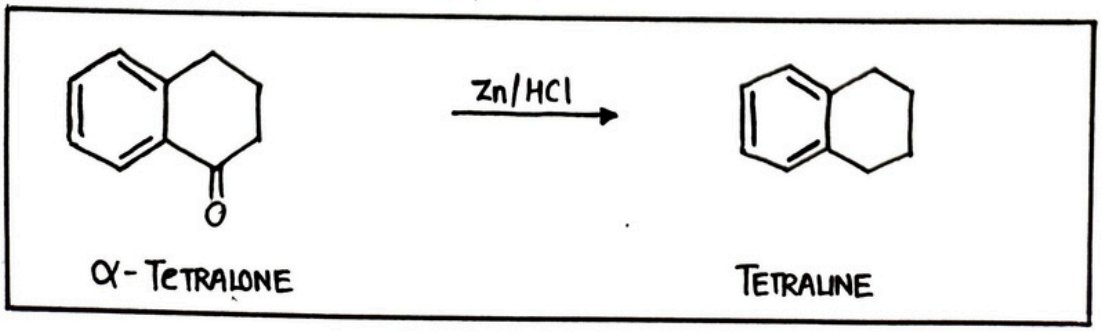
- **STEP - III (Ring Closure Reaction)**

4- Phenyl Butanoic Acid is heated with conc. H_2SO_4 to form α- tetralone.



- **STEP - IV (Clemmensen Reduction)**

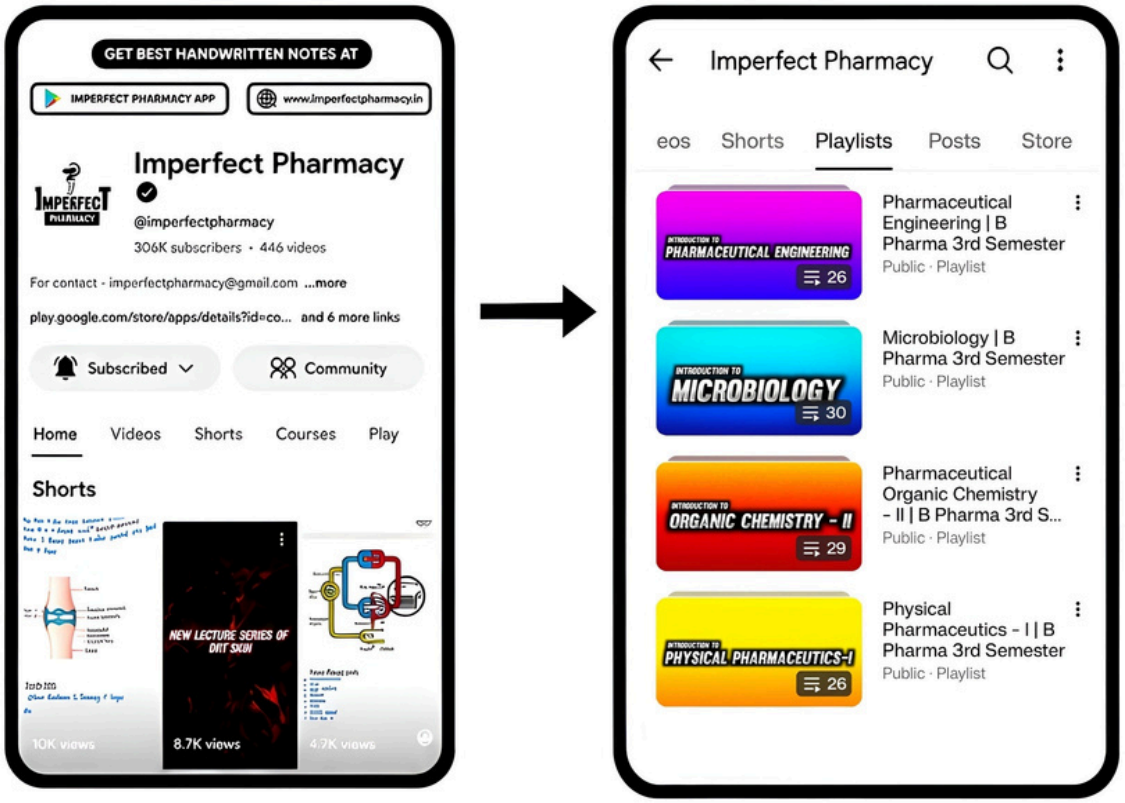
α Tetralone heated with zinc & HCl to give Tetraline



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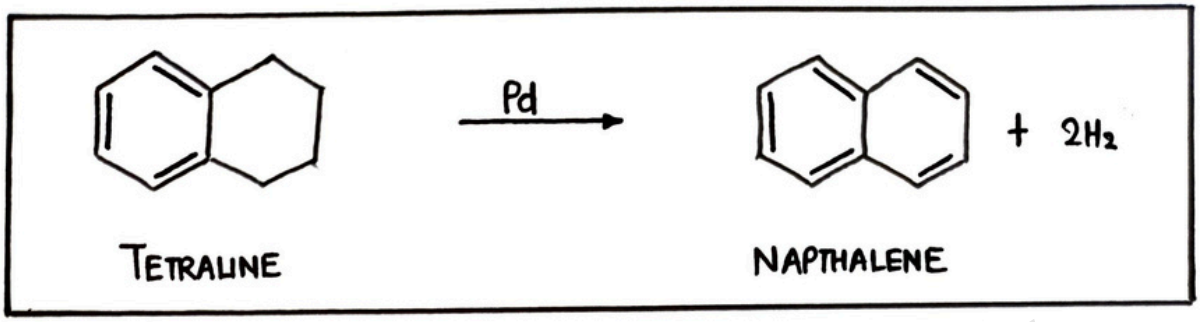
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STEP - V (Aromatization)

Tetralin is heated with Palladium to give Napthalene.

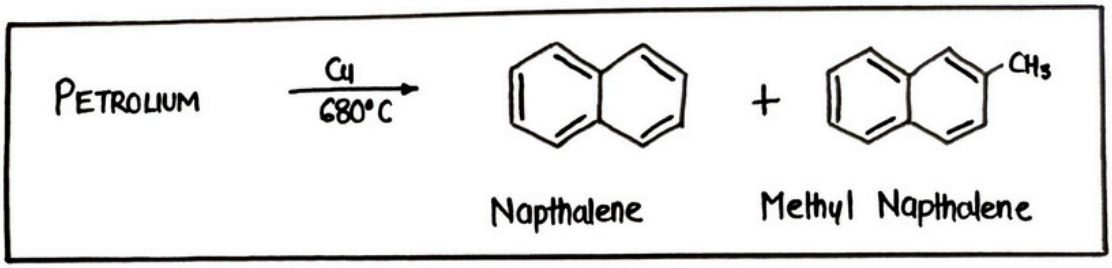


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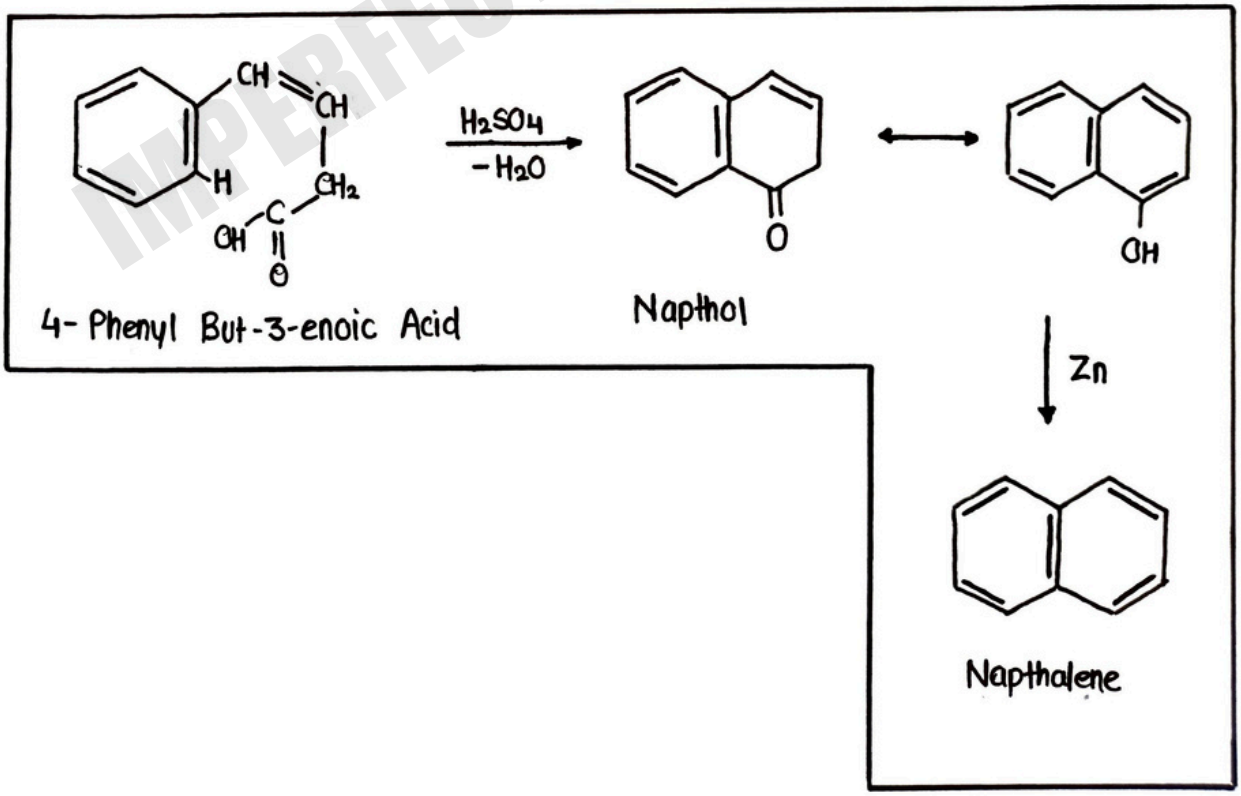
② From Petrolium

When Petrolium fraction passed over Copper (Cu) catalyst at 680°C, it leads to the formation of Napthalene & Methyl napthalene.



③ From 4 Phenyl But-3enoic Acid

When 4 Phenyl But-3-enoic Acid is heated with concentrated H_2SO_4 it gives napthol which on further distillation with zinc dust yellds Napthalene.

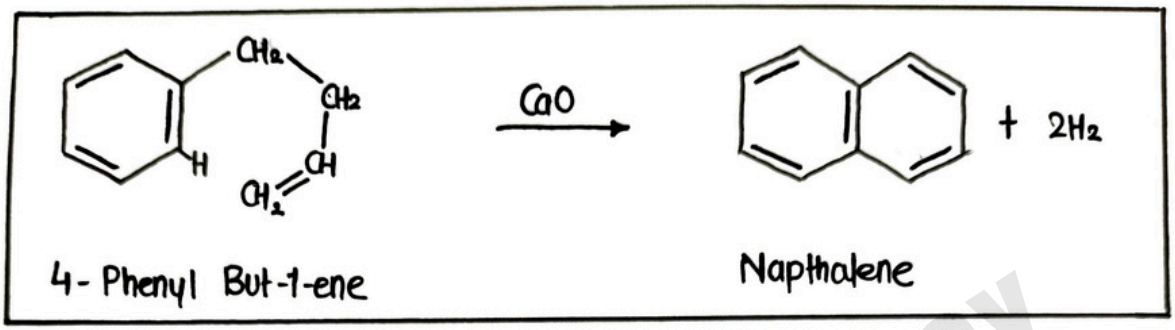


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④ From 4 Phenyl But-1-ene

Whene 4-Phenyl But-1-ene is passed over Red Hot Calcium Oxide, it yeilds the formation of Napthalene.



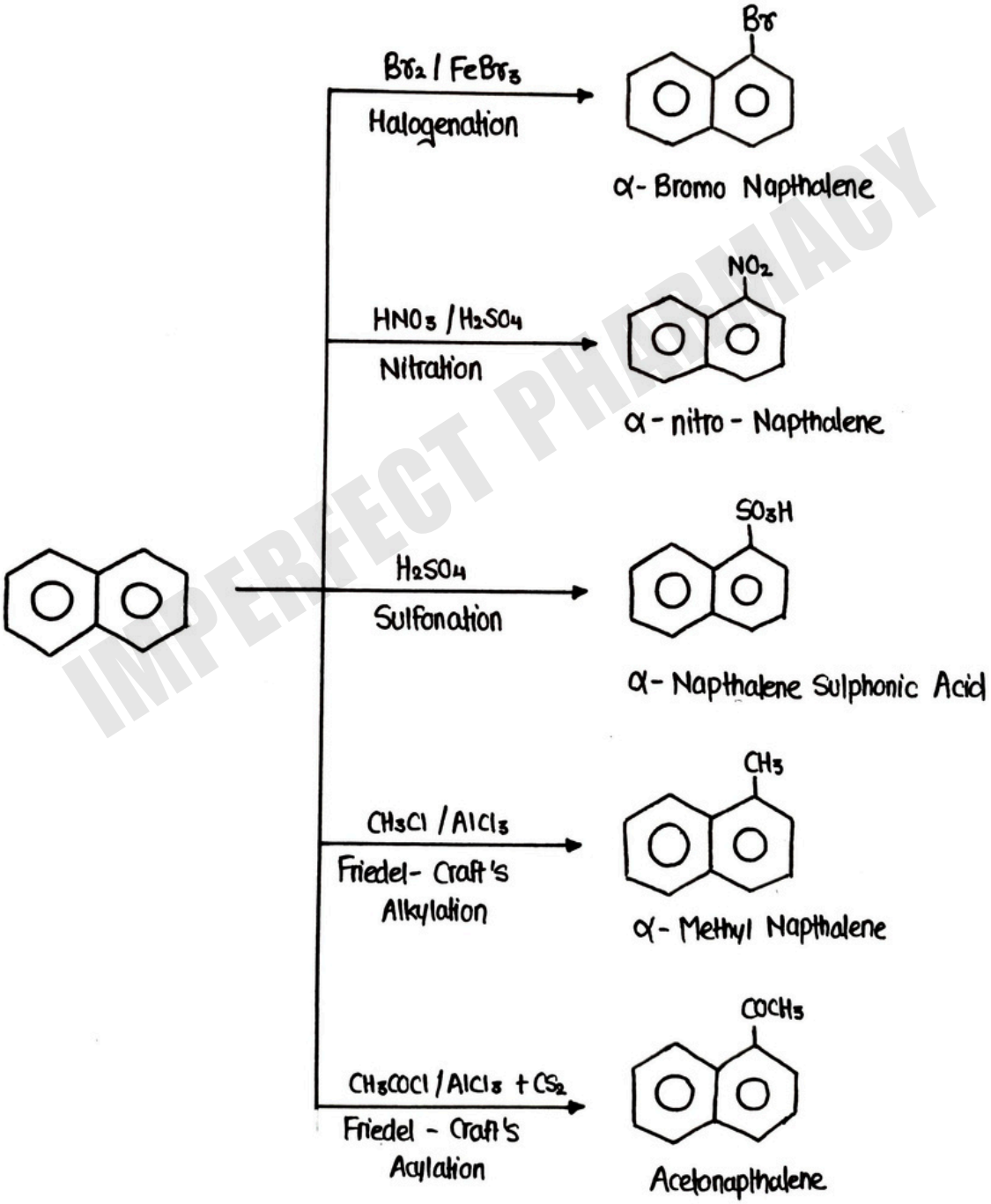
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CHEMICAL REACTIONS

Napthalene undergoes the following chemical reactions :

① ELECTROPHILIC SUBSTITUTION REACTION

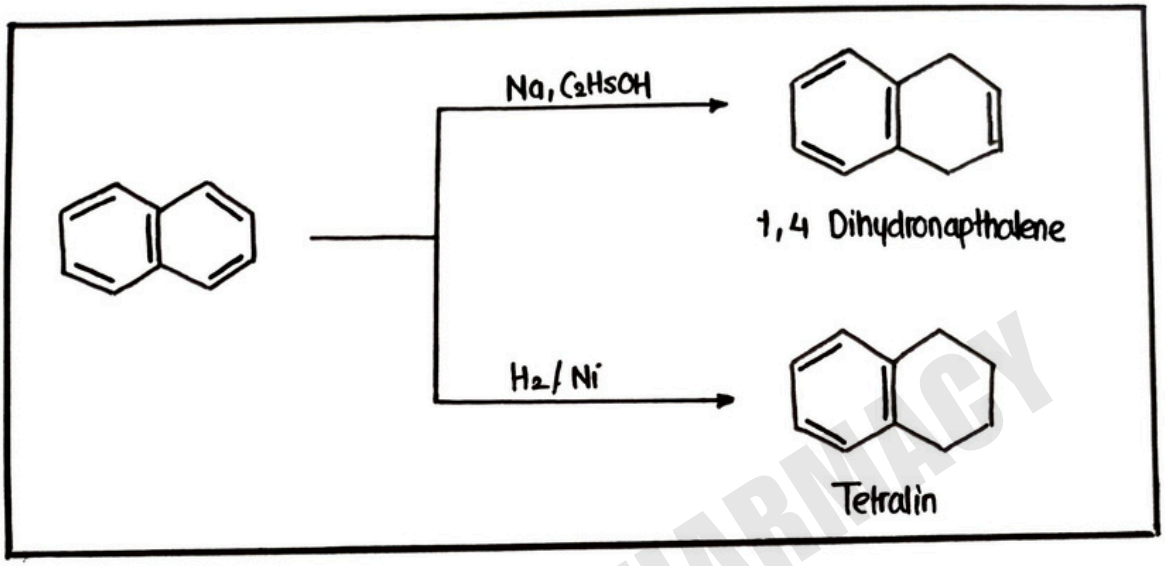


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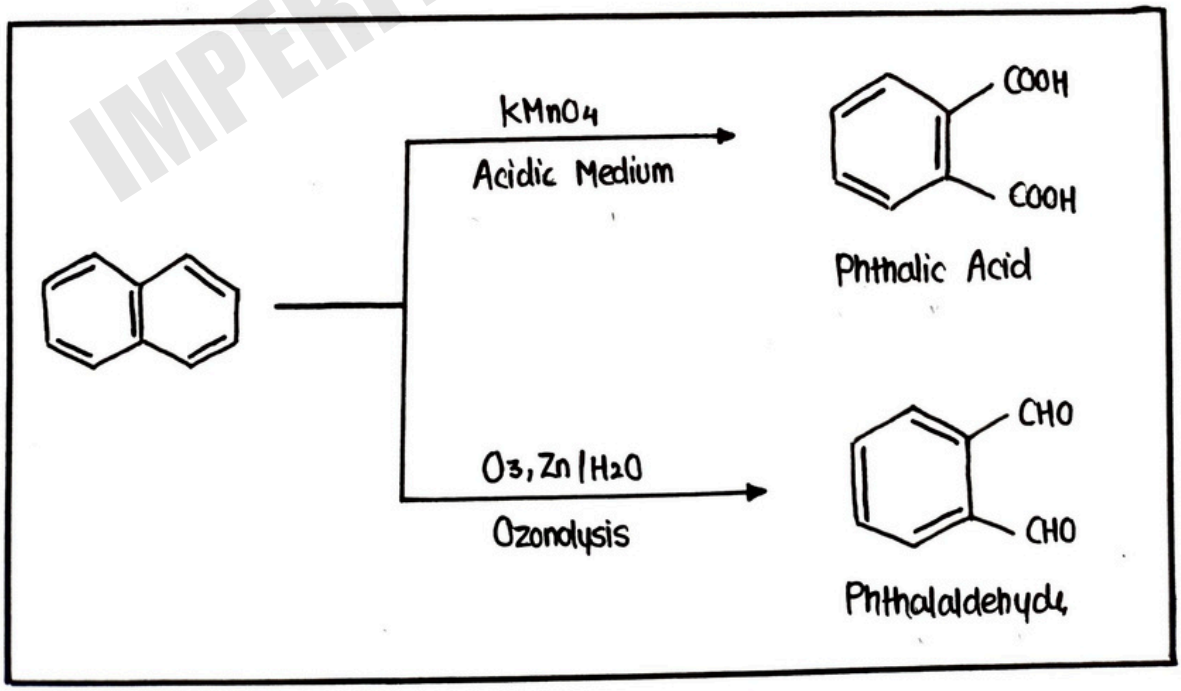
② Reduction

The reduction reaction in benzenes depends upon different reducing agent.



③ Oxidation

Oxidation in napthalene depends upon different oxidizing agent.



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Medicinal Uses

- It is used as Antiseptic Agent.
- It is used as Preservative.
- It is effectively used in Eye troubles.
- It is effective in treating the symptoms of cold, Fever, Gonorrhoea etc.
- It is used in synthesis of various dyes.
- It is also used in synthesis of various drugs like Nadoxolol.

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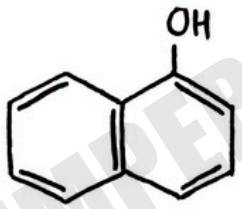
NAPTHALENE DERIVATIVES

There are various Napthalene derivatives given below :

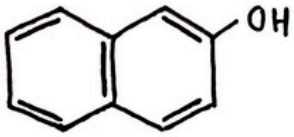
- Napthols
- Napthylamines
- Naptholic Acid
- Napthoquinones

NAPTHOLS

- It is the hydroxyl derivative of Napthalene.
- It is the precursor for various drug synthesis.
- α-Napthol is the main ingredient of Molisch Reagent.
- β-Napthols are used as antiseptic for skin diseases.



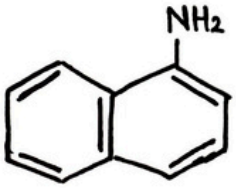
α-Napthol



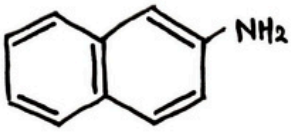
β-Napthol

NAPTHYLAMINES

- Napthylamine is a white or yellow crystalline substance.
- It is used in dye manufacturing, rubber, pesticides & various pharmaceuticals.



α-Napthylamine



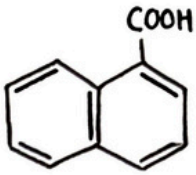
β-Napthylamine

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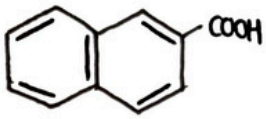
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③ NAPTHOIC ACID

- They are aromatic carboxylic acids used in various organic synthesis.
- It is used as intermediates in production of dyes & pharmaceuticals.



α- Naptholic Acid

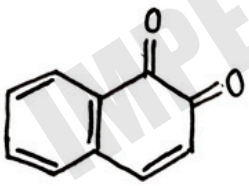


B-Naptholic Acid.

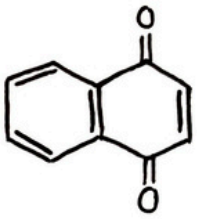
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④ NAPTHOQUINONES

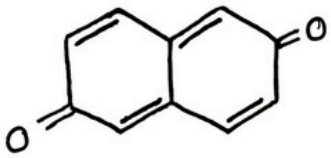
- These are compounds that contain a napthalene ring with two ketone Groups.
- They have Antimicrobial & Antifungal Properties.
- They are used in various dyes & drug development.



1,2 Napthoquinone



1,4 Napthoquinone

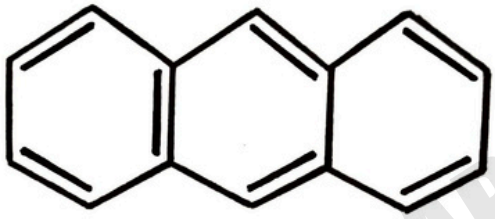


2,6 Napthoquinone

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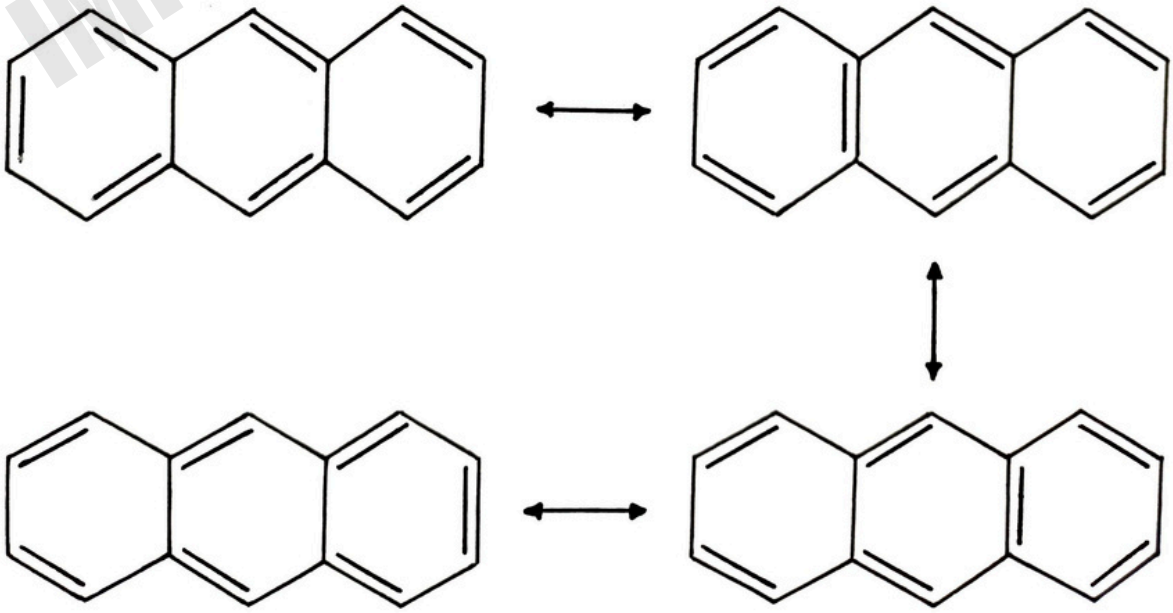
ANTHRACENE

- Anthracene is a solid polycyclic Aromatic hydrocarbon consist of 3 linear fused benzene ring.
- It is a member of Fused ring hydrocarbons under the class of Polynuclear Hydrocarbons.



RESONANCE IN ANTHRACENE

- Anthracene undergoes delocalization to give 4 resonance structure :
- The presence of resonance makes Anthracene stable.

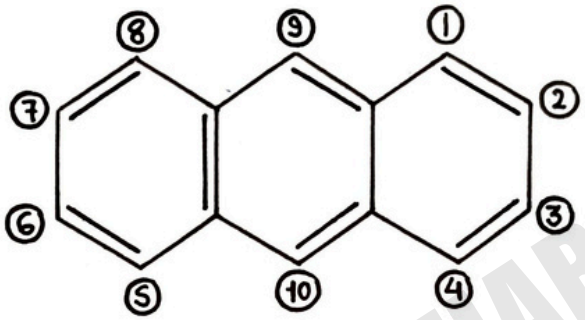


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Chemical Properties Of Anthracene

- The chemical formula for Anthracene is $C_{14}H_{10}$ & molecular weight is 178 g/mol.
- All the carbons in Anthracene are sp^2 hybridized.
- Anthracene is more reactive than benzene because resonance energy of Anthracene is 84 kcal/mol which is less than thrice of benzene (36 kcal/mol).



- Position no 1, 4, 5, 8 are similar known as : α Position
- Position no 2, 3, 6, 7 are similar known as : β Position
- Position no 9, 10 are similar and known as : γ Position
- C_1-C_2 bond have more double bond character & hence shorter bond length (1.37 Å).
- C_2-C_3 bond have more single bond character & hence longer bond length (1.42 Å).

Physical Properties

- It is a colourless solid.
- Its Melting Point is 218°C
- Its boiling point is 340°C
- It is insoluble in water but soluble in organic solvent.

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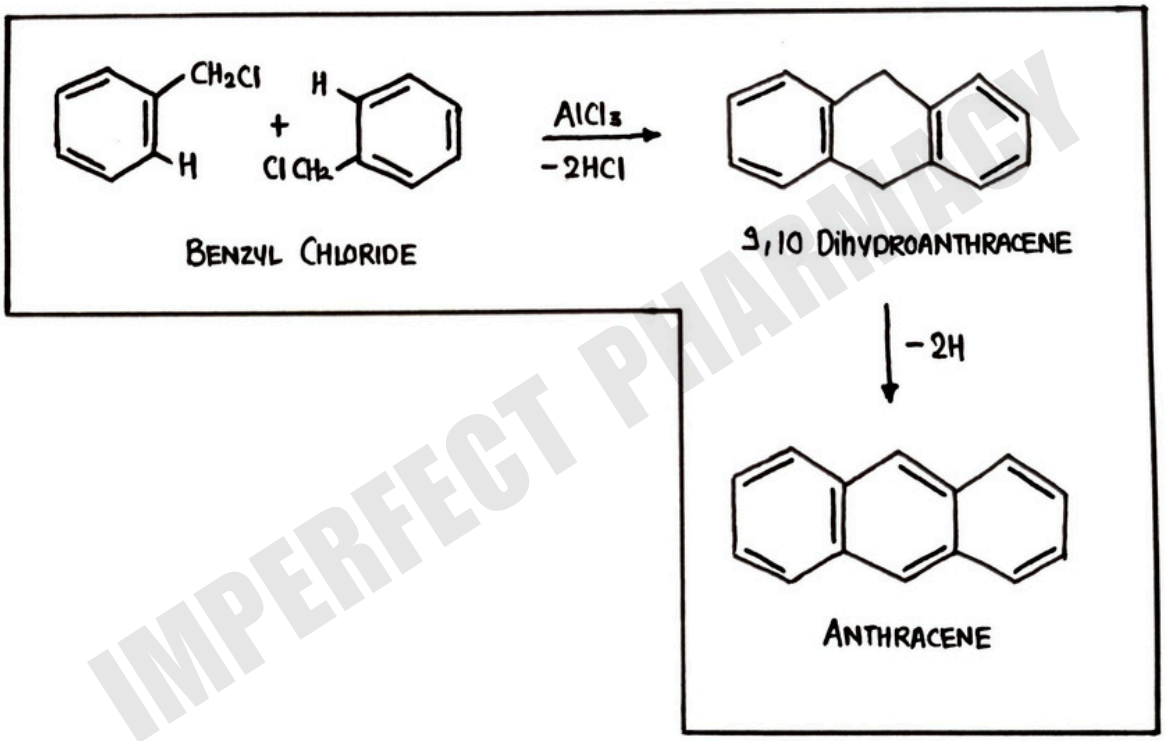
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SYNTHESIS

Anthracene can be synthesized / prepared by following methods :

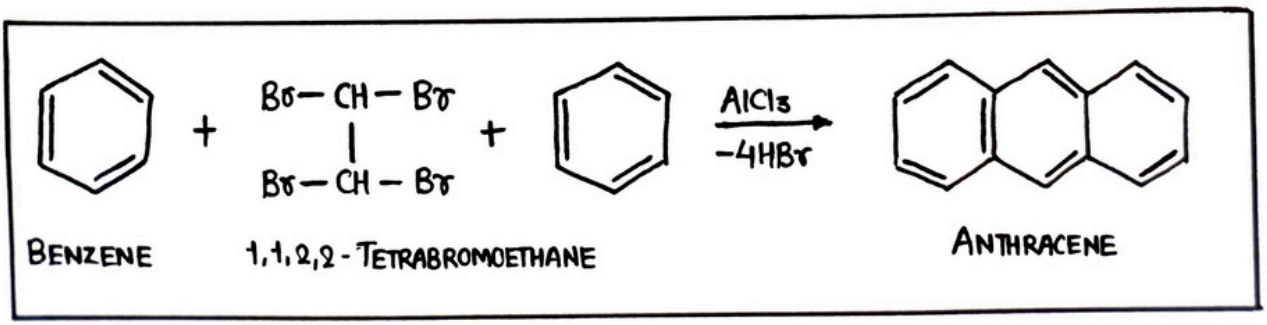
① FRIEDEL CRAFTS REACTION

Two molecules of Benzyl chloride reacts to form 9,10 Dihydroanthracene which on further reaction losses two Hydrogen to form Anthracene.



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- Anthracene can also be prepared by reaction of 2 benzene molecules with 1,1,2,2 - Tetrabromo Ethane.

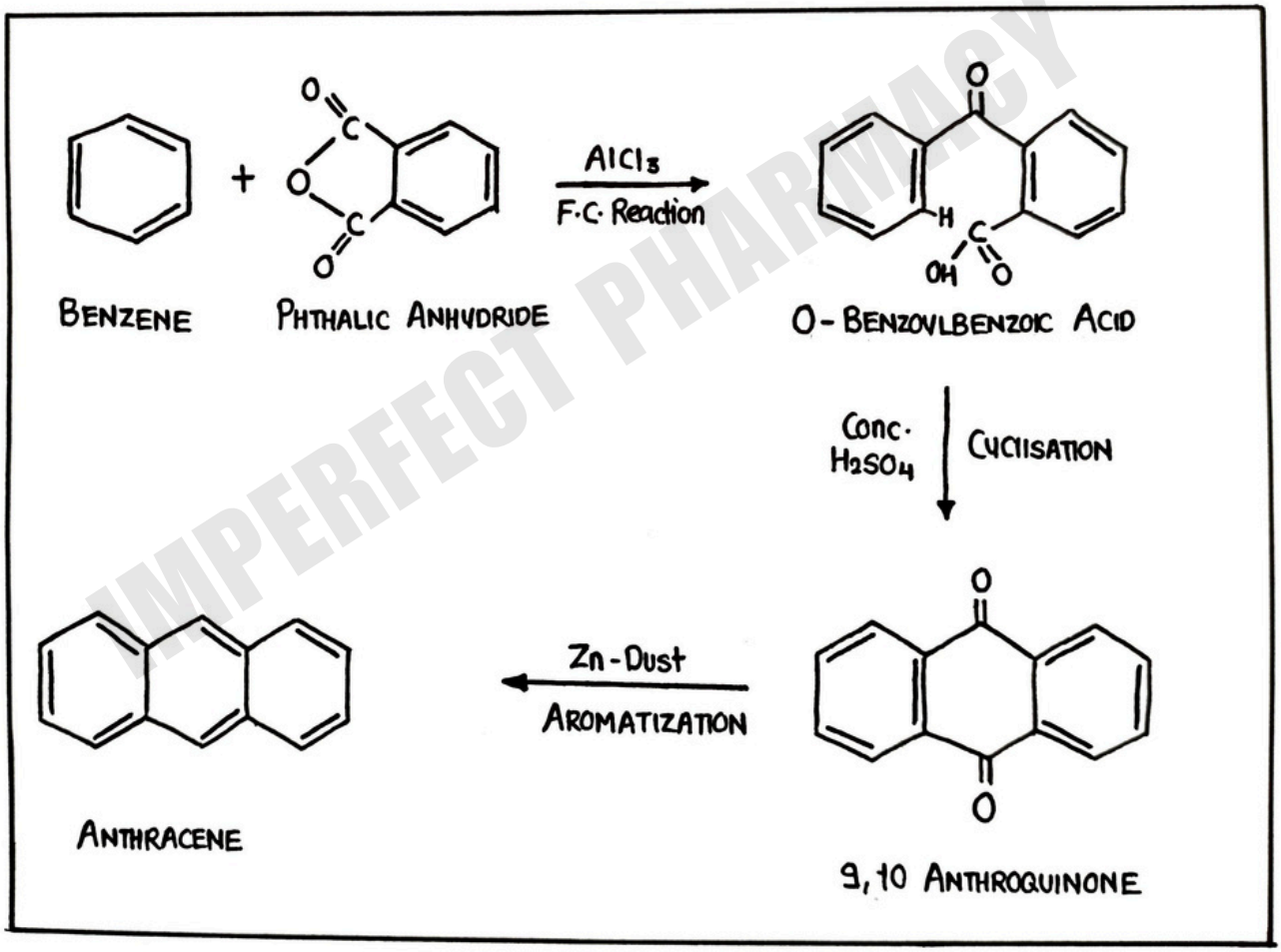


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② By Haworth Synthesis

In Haworth synthesis, Benzene reacts with Pthalic Anhydride in the presence of Aluminium Chloride to form O- Benzoyl Benzoic Acid, then O- Benzoyl Benzoic Acid heated with concentrated Sulphuric Acid (H_2SO_4) to give 9,10- anthraquinone.

Now this 9,10- anthraquinone on further distillation with zinc dust yeilds anthracene.

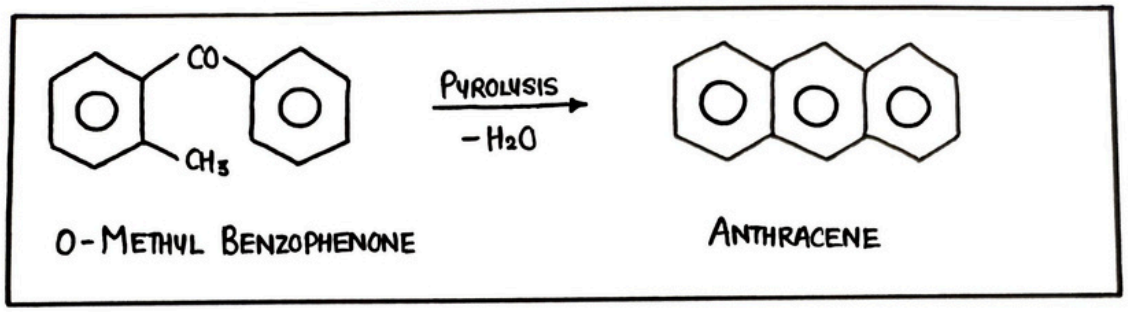


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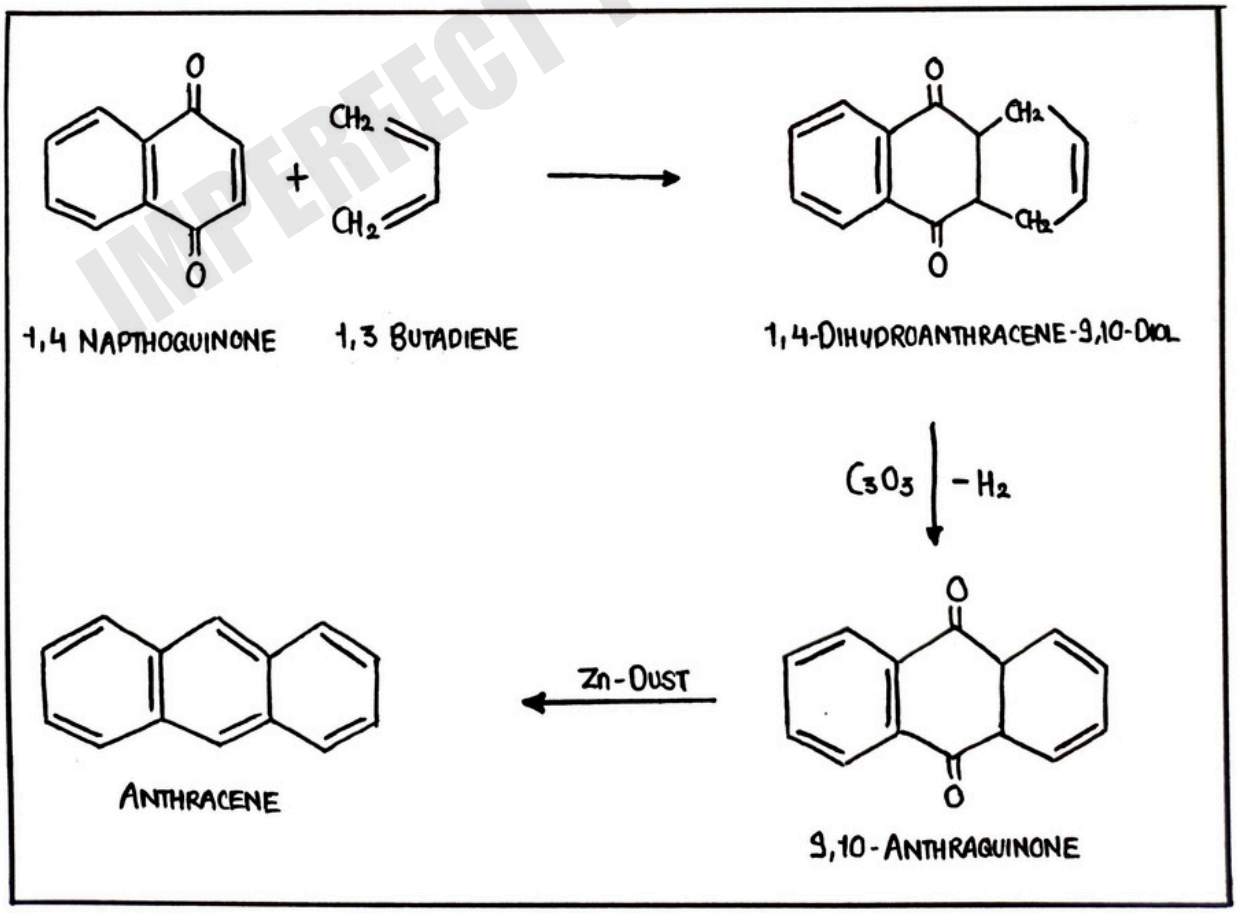
③ Elbs Reaction

Anthracene is prepared by pyrolysis of O-Methyl benzophenone.



④ By Diels-Alder Reaction

When 1,4-napthaquinone reacts with 1,3-butadiene it forms 1,4-Dihydroanthracene-9,10-diol which on oxidation with Chromium Trioxide gives 9,10-anthraquinone which on further distillation with zinc dust gives anthracene.



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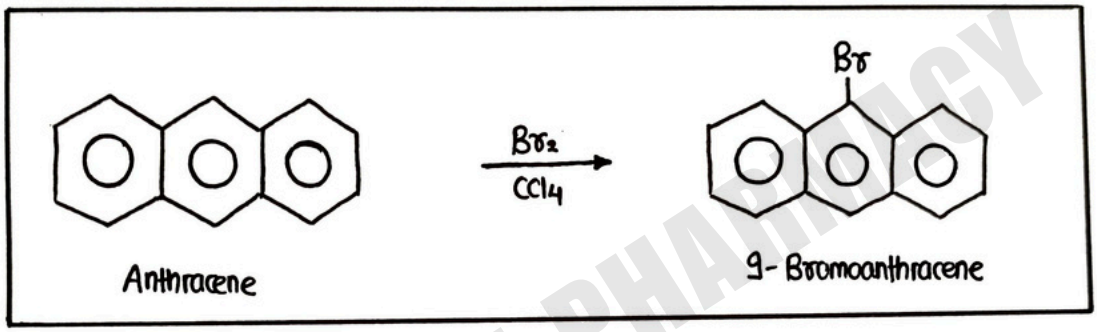
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CHEMICAL REACTIONS

Anthracene undergoes following chemical reactions :

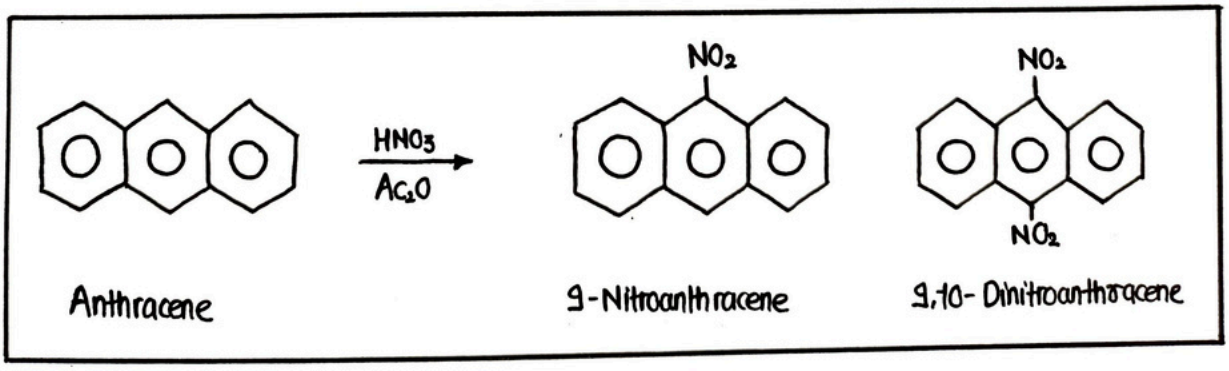
① HALOGENATION

Anthracene reacts with Halogens (Cl / Br) in the presence of CCl_4 to form 9- Bromoanthracene.



② NITRATION

Anthracene reacts with concentrated HNO_3 in the presence of Acetic Anhydride to form mixture of 9- Nitroanthracene & 9,10- Dinitroanthracene.

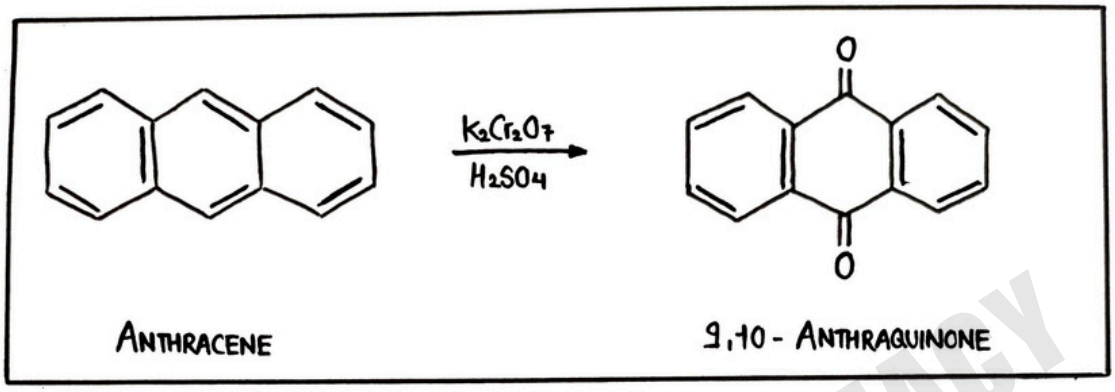


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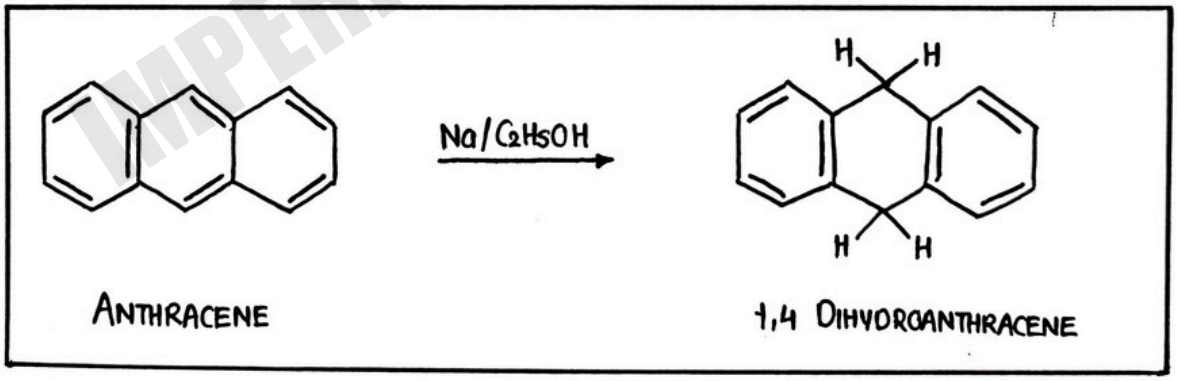
③ OXIDATION

Anthracene undergoes oxidation in the presence of strong oxidizing Agent like Potassium Dichromate ($K_2Cr_2O_7$) to give 9,10 Anthraquinone.



④ REDUCTION

Anthracene undergoes reduction in the presence of sodium & ethanol to form 9,10 - Dihydroanthracene.



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MEDICINAL USES

- It is used as Wood Preservatives.
- It is used as insecticides.
- It is also used Antifungal & Antiviral Agent.
- Used as constituents of drugs like Senna, Cascara etc.
- It is also used in manufacturing of various dyes.

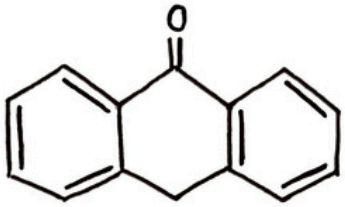
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ANTHRACENE DERIVATIVES

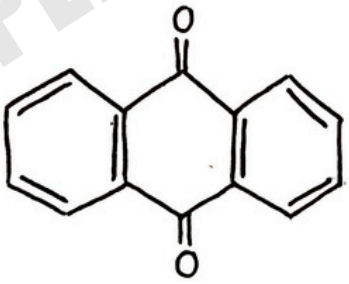
There are various Anthracene derivatives as follows :-

① ANTHRONE



- It is used for determination of Carbohydrates.
- It is used to detect & quantify sugars in different samples.
- It is also used as Laxative

② ANTHRAQUINONE



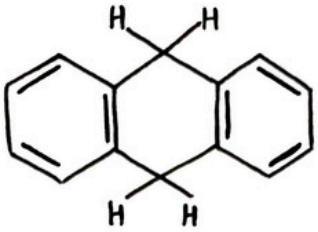
USES

- It is used for manufacturing various dyes.
- It is used as herbicides & pesticides.
- It serves as building block for the synthesis of various organic compounds.

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③ DIHYDRO- ANTHRACENE



USES

- It is used in Photographic techniques.
- It serves as building blocks in synthesis of various organic compounds.

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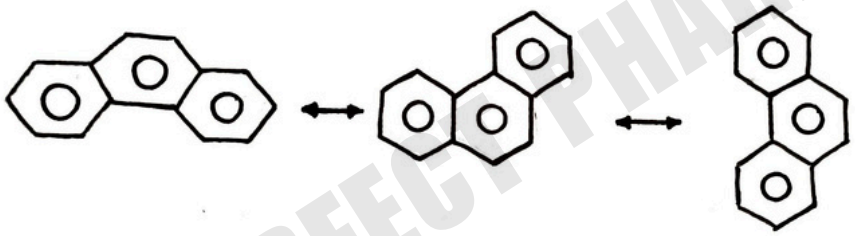
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PHENANTHERENE

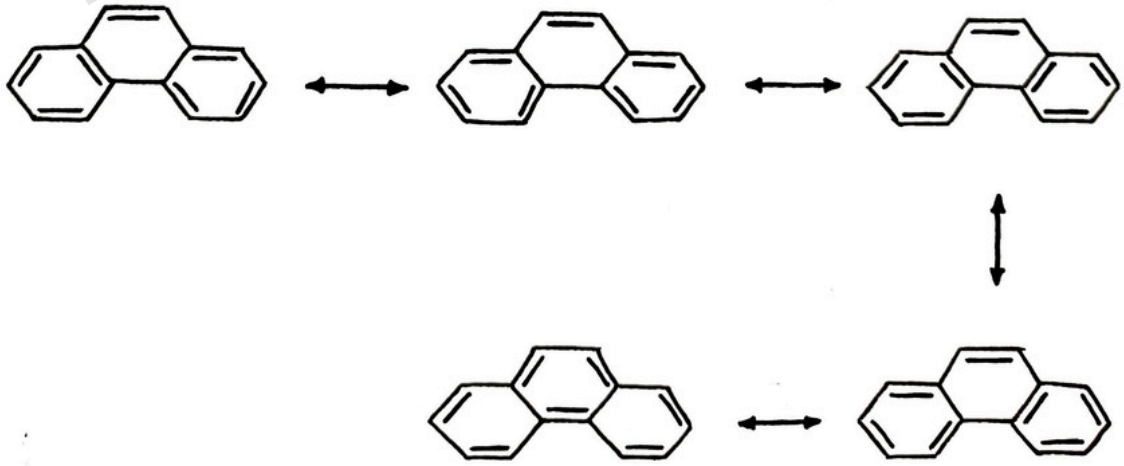
- Phenantherene is a polycyclic aromatic hydrocarbon composed of three fused benzene rings.
- It is a structural isomer of Anthracene.
- It is a member of Fused Ring Hydrocarbons under the class of Polynuclear Hydrocarbons.

REPRESENTATION

Phenantherene can be represent in following 3 ways :



RESONANCE IN PHENANTHERENE

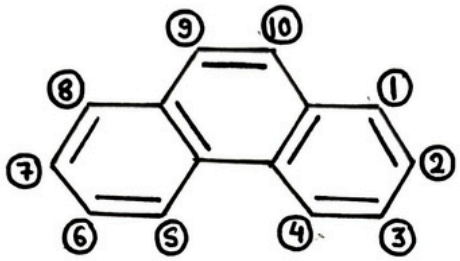


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Chemical Properties

- Its chemical formula is $C_{14}H_{10}$.
- It is an isomer of Anthracene.
- Its molecular weight is 178 g/mol.
- All the carbons of Phenantherene are sp^2 hybridized.
- It follows Huckle Rule.



Physical Properties

- Phenanthesene is a colourless crystalline solid.
- Its Melting Point is 101°C.
- Its boiling point is 340°C.
- Its insoluble in water but soluble in organic solvents.

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SYNTHESIS

Phenanthrene can be synthesized by following reactions :

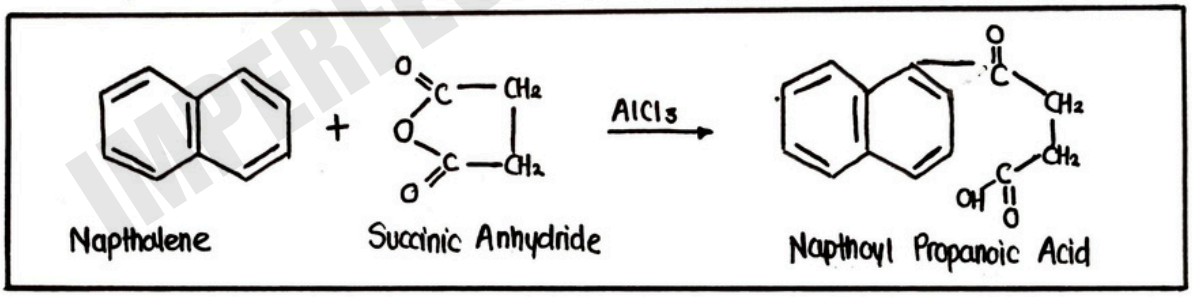
① HAWORTH SYNTHESIS

It involves 5- step chemical reaction :

- Friedel Craft Acylation
- Clemmensen Reduction
- Ring Closure Reaction
- Clemmensen Reduction
- Aromatization

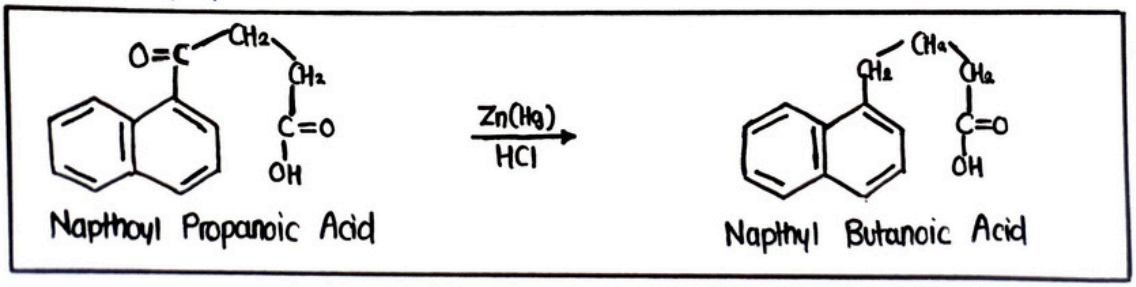
● Friedel Craft Acylation

Napthalene reacts with succinic anhydride in the presence of $AlCl_3$ to form Napthoyl Propanoic Acid.



● Clemmensen Reduction

Napthyl Propanoic Acid reacts with amalgated zinc in the presence of HCl to form Napthyl Butanoic Acid.

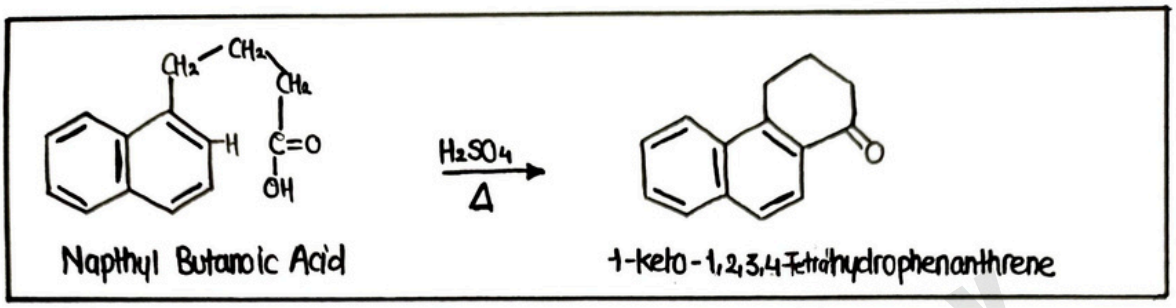


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Cyclisation

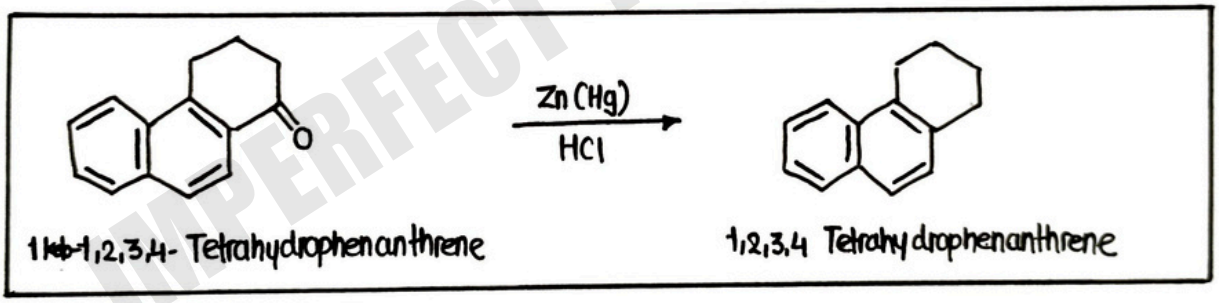
Napthyl Butanoic Acid heated in the presence of H_2SO_4 to give 1-keto,-1,2,3,4,-~~Dihydrophenanthrene~~. Tetrahydrophenanthrene.



Napthyl Butanoic Acid → 1-keto-1,2,3,4-Tetrahydrophenanthrene

Clemmensen Reduction

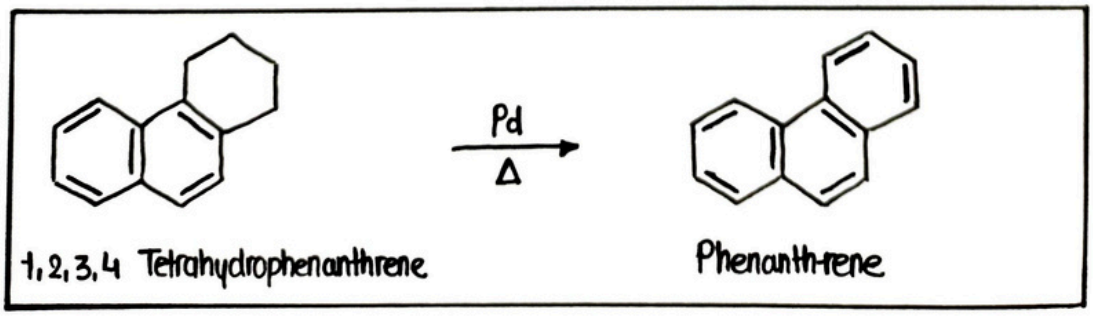
1-keto-1,2,3,4 Tetrahydrophenanthrene reacts with Amalgamated zinc in the presence of HCl to give 1,2,3,4 Tetrahydrophenanthrene.



1-keto-1,2,3,4-Tetrahydrophenanthrene → 1,2,3,4 Tetrahydrophenanthrene

Aromatisation

1,2,3,4 Tetrahydrophenanthrene heated with Palladium to give Phenanthrene.



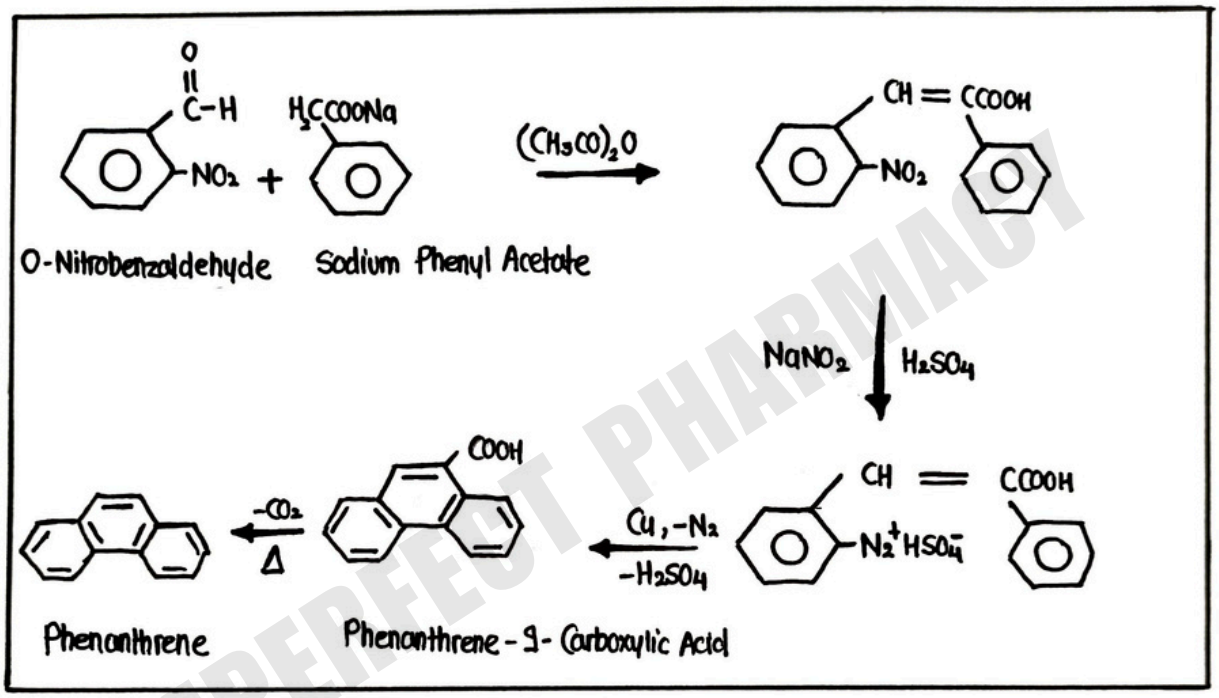
1,2,3,4 Tetrahydrophenanthrene → Phenanthrene

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② PSCHORR SYNTHESIS

O- Nitrobenzaldehyde is heated with Sodium phenyl acetate in the presence of acetic anhydride to give Phenyl -o- nitro cinnamic acid which undergoes diazotization which on further reaction gives Phenanthrene- 9- carboxylic acid which on heating give Phenanthrene.



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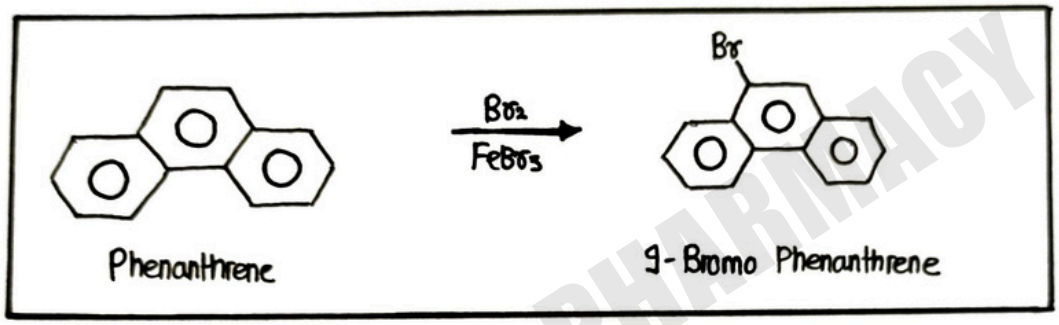
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CHEMICAL REACTIONS

Phenanthre undergoes following chemical reactions :

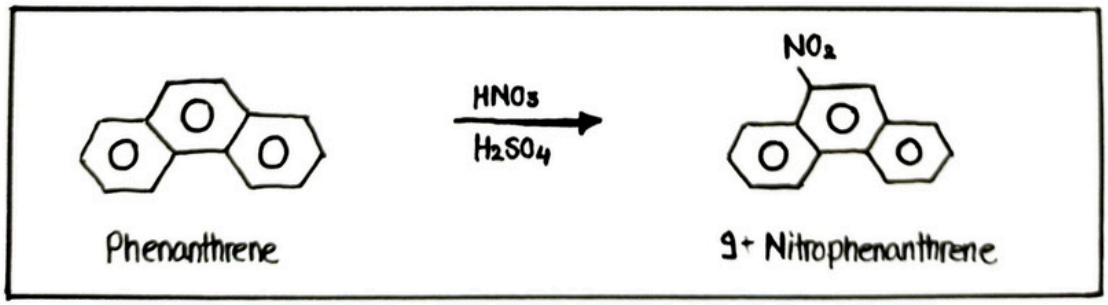
① HALOGENATION

Phenanthrene reacts with Bromine in the presence of $FeBr_3$ to give 9- bromophenanthrene.



② NITRATION

Phenanthrene undergoes Nitration with concentrated Nitric Acid & Sulphuric Acid to give 9- Nitrophenanthrene.

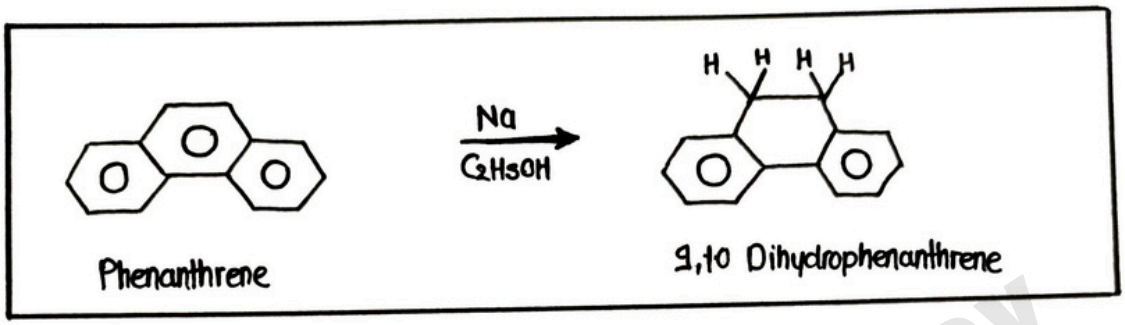


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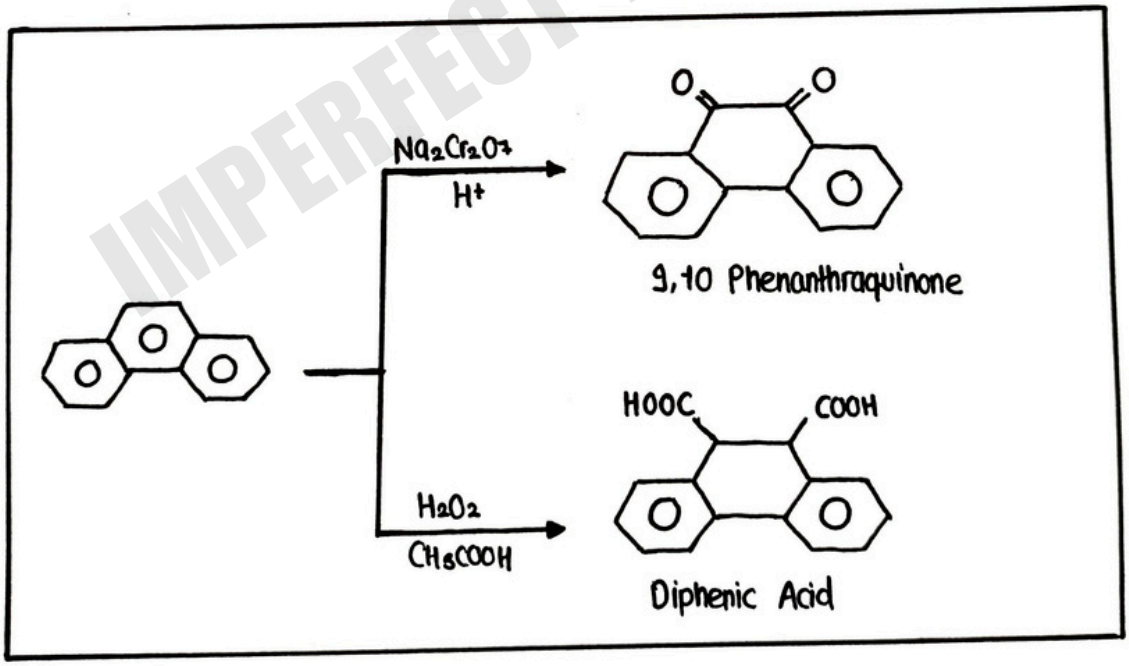
③ REDUCTION

Phenanthrene undergoes reduction in the presence of sodium & ethanol to form 9,10 Dihydrophenanthrene.



④ OXIDATION

Oxidation product of Phenanthrene depends on oxidizing agent.



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Uses Of Phenanthrene

- It is used in the manufacturing of Pesticides & dyes..
- It is used in the production of Jet Fuel.
- It can be used in the synthesis of Morphine, Caffeine etc.
- It is used as Anticancer Agent.
- It is used as Antimalarial drugs.
- It is also used as Antioxidant & Antimicrobial

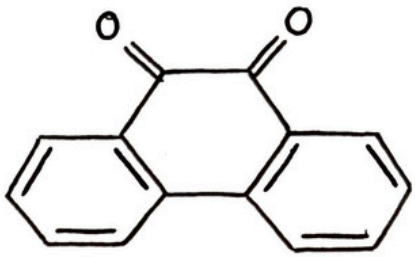
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DERIVATIVES

The most important derivative of Phenanthrene is Phenanthraquinone.

PHENANTHRAQUINONE



USES

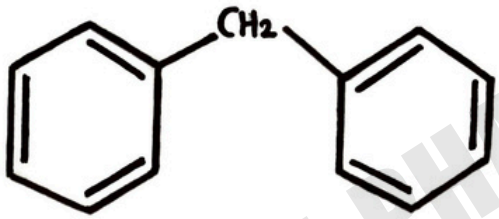
- It is used in the synthesis of various dyes.
- It is used as precursor for synthesis of various organic compounds.
- It is also used as Antimicrobial & Anticancer Agent.
- It can also be used as redox indicator in Analytical Chemistry.

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DIPHENYL METHANE

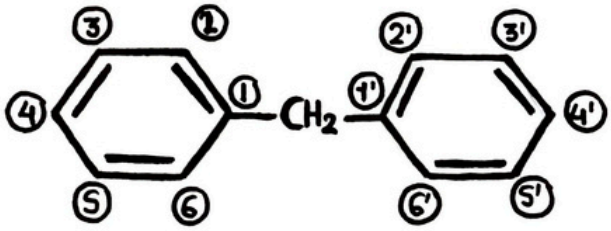
- Diphenyl Methane is an organic compound consist of Methane where two hydrogen atoms are replaced by two hydrogen ~~atoms~~ Phenyl Groups.
- It is a member of isolated ring hydrocarbons under the class of Polynuclear Hydrocarbons.
- It is also known as Benzhydryl.



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PHYSICAL & CHEMICAL PROPERTIES

- Its Chemical formula is $C_{13}H_{12}$.
- It is crystalline solid.
- Its Meting Point is 22-24°C.
- Is boiling point is 264°C.
- It is insoluble in water & soluble in organic solvents.



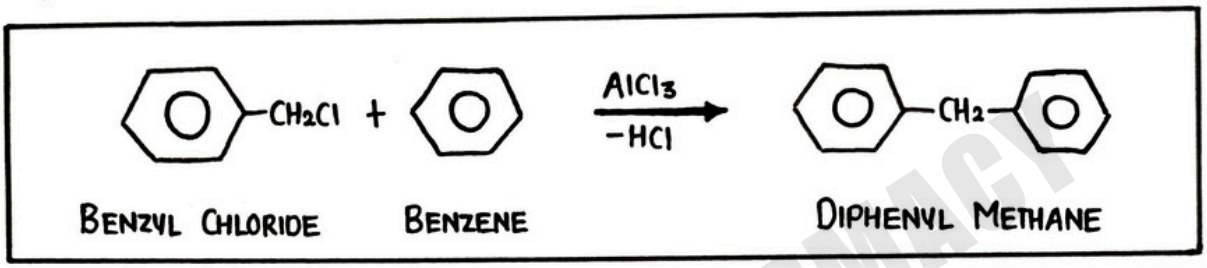
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SYNTHESIS

Diphenyl Methane is synthesized by following methods :

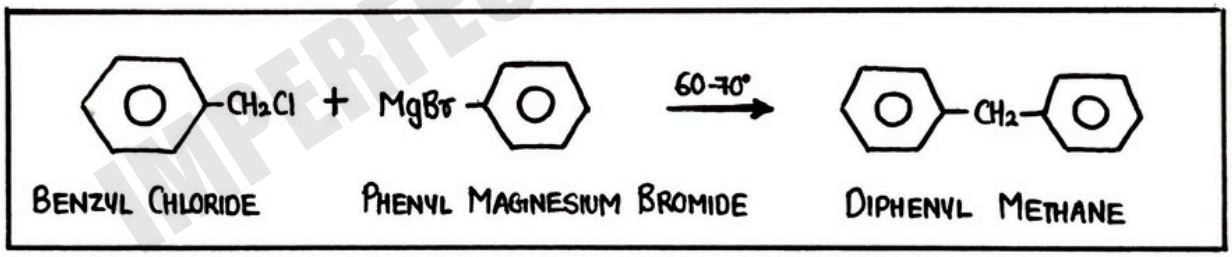
① FRIEDEL CRAFT'S CONDENSATION

When Benzyl Chloride reacts with benzene in the presence of $AlCl_3$, it yeilds Diphenylmethane.



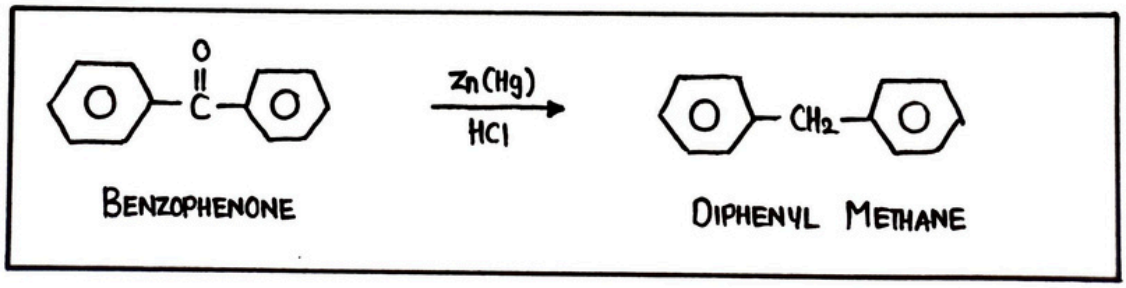
② FROM GRIGNARD REAGENT

Phenyl Magnesium Bromide reacts with Benzyl Chloride to give Diphenyl Methane.



③ WOLF - KISHNER REACTION

Benzophenone reduced to Diphenyl methane through Clemmensen Reduction.



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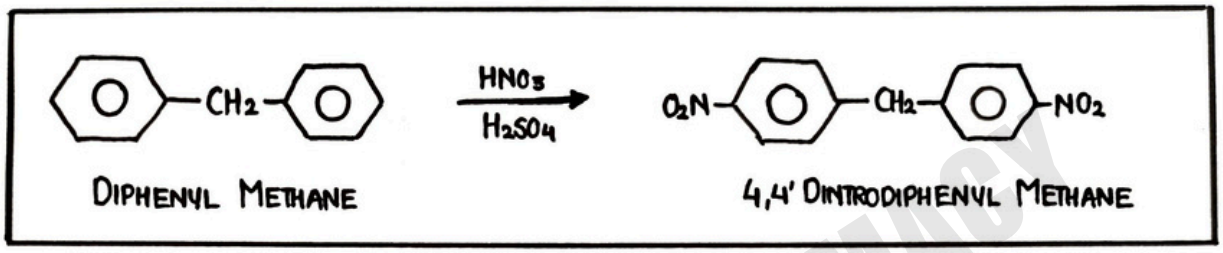
VIDEOS : IMPERFECT PHARMACY

CHEMICAL REACTIONS

Diphenyl Methane undergoes following chemical reactions :

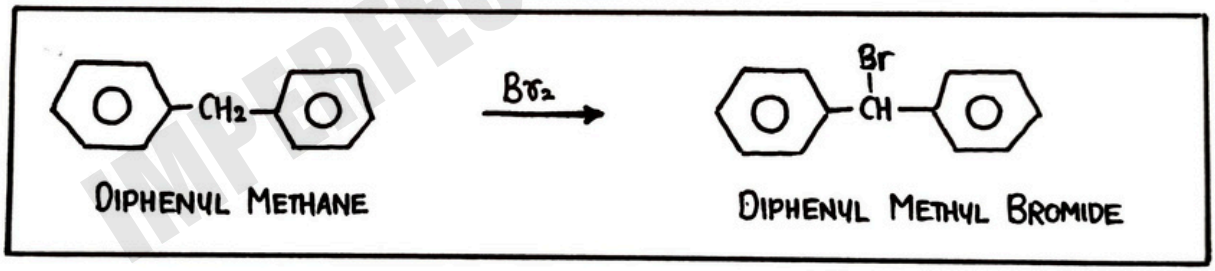
① NITRATION

Diphenyl Methane reacts with conc. HNO_3 to form 4,4' Dinitro-Diphenyl Methane.



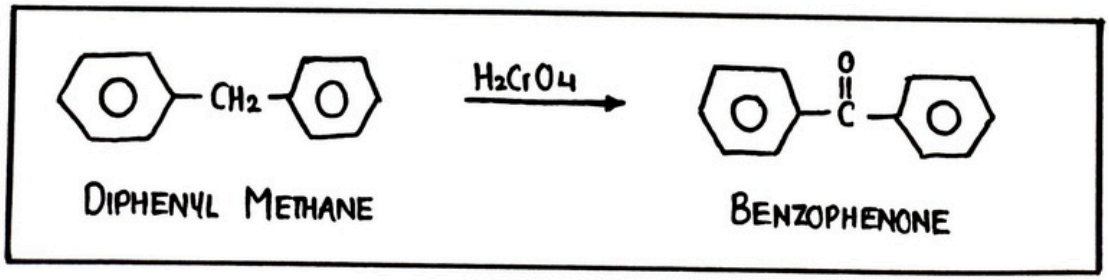
② BROMINATION

Diphenyl Methane reacts with Bromine to form Diphenyl Methyl Bromide.



③ OXIDATION

Diphenyl Methane undergoes oxidation to form Benzophenone



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Uses Of Diphenyl Methane

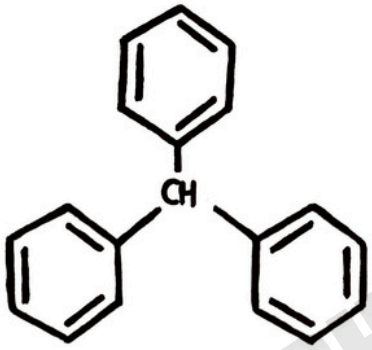
- It is used for manufacturing of soaps, Pesticides & Insecticides.
- It is also used as Jet Fuels.
- It is used for synthesis of various dyes.
- It is used as Antiulcer Agent.
- It is used as Antihistamine Agent.
- It is also used as Anti-Anxiety Drugs.

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TRIPHENYL METHANE

- Triphenyl Methane is an organic compound in which 3 hydrogen atoms of Methane are replaced by Phenyl Groups.
- It is a part of isolated ring hydrocarbons under the class of Polynuclear Hydrocarbons.



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PHYSICAL & CHEMICAL PROPERTIES

- Its chemical formula is $C_{19}H_{16}$.
- It is a colourless solid.
- Its Melting Point is 92-94°C.
- Its boiling point is 359°C
- It is insoluble in water but soluble in organic solvents.

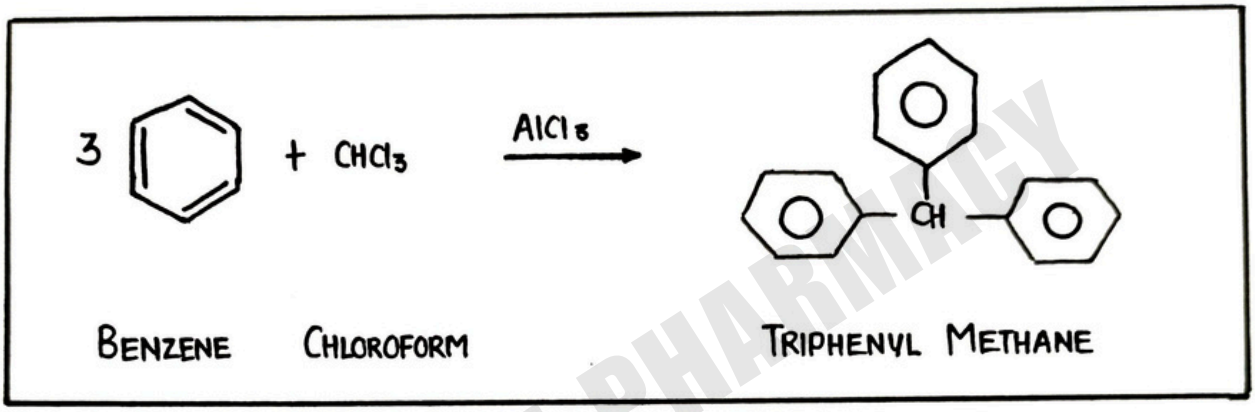
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SYNTHESIS

Triphenyl Methane is synthesized by following reactions :

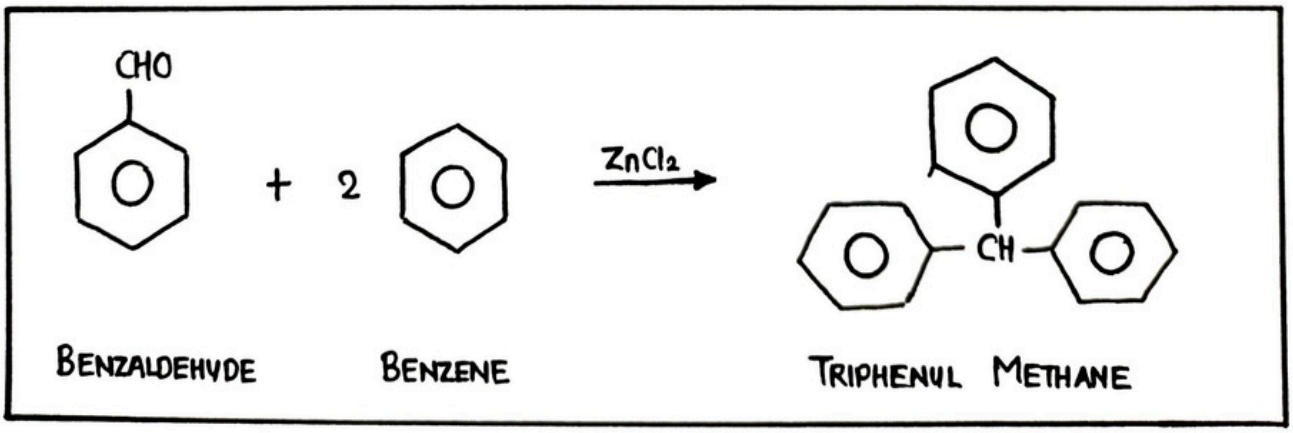
① FRIEDEL CRAFT'S REACTION

3 Molecules of Benzene reacts with Chloroform to form Triphenyl Methane.



② FROM BENZALDEHYDE

Benzaldehyde reacts with 2 molecules of benzene in the presence of $ZnCl_2$ to yeild Triphenyl Methane

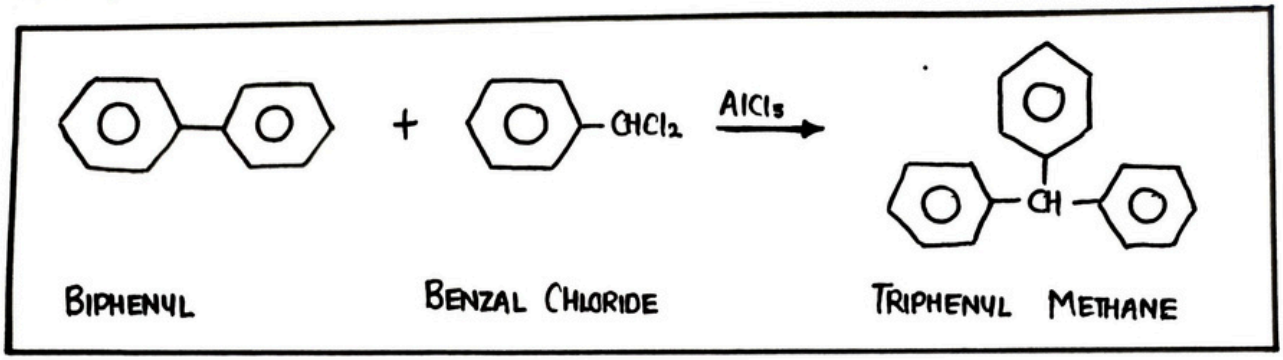


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③ FROM BIPHENYL

Biphenyl reacts with Benzal Chloride to form Triphenyl Methane.



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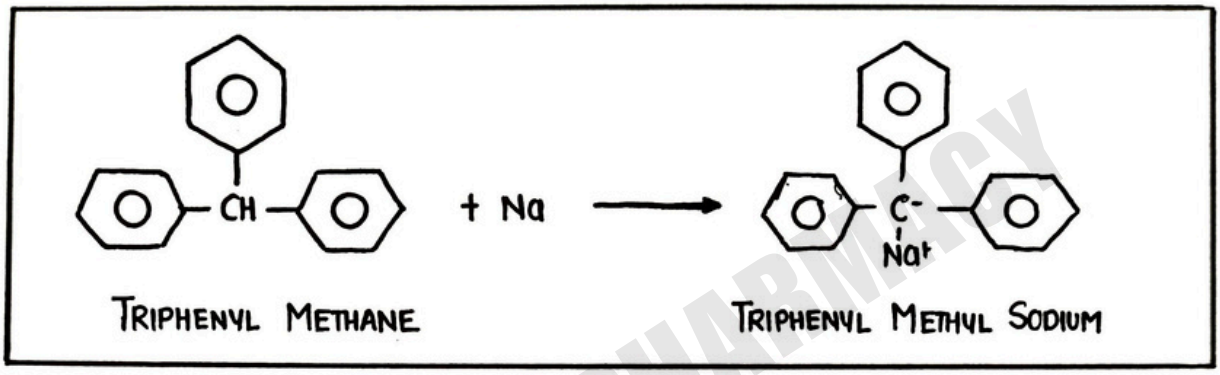
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CHEMICAL REACTIONS

Triphenyl Methane undergoes following chemical reactions :

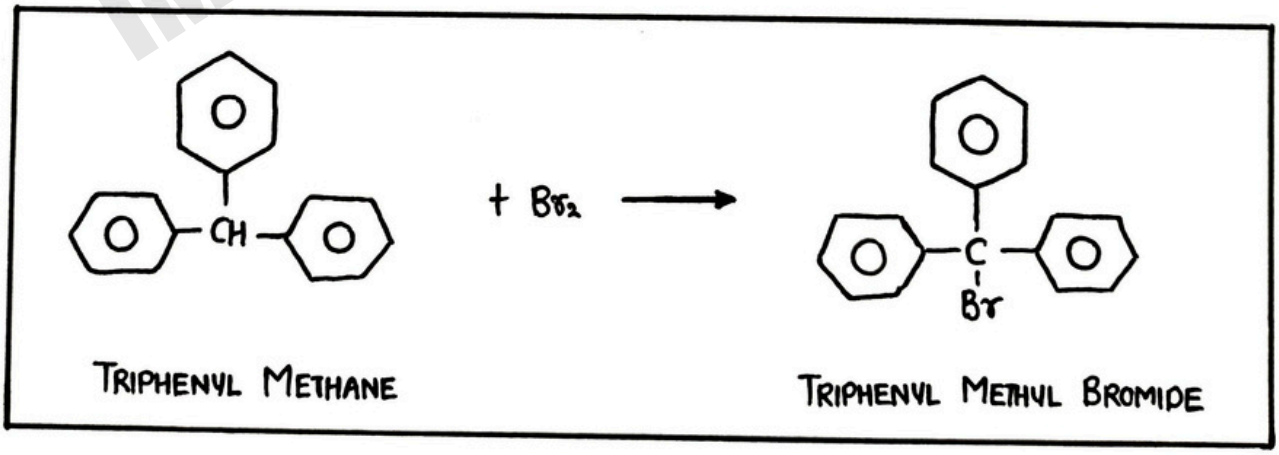
① REACTION WITH SODIUM

Triphenyl Methane reacts with sodium to give Triphenyl Methyl Sodium.



② BROMINATION

Triphenyl Methane reacts with Bromine to yeild Triphenyl Methyl Bromide.

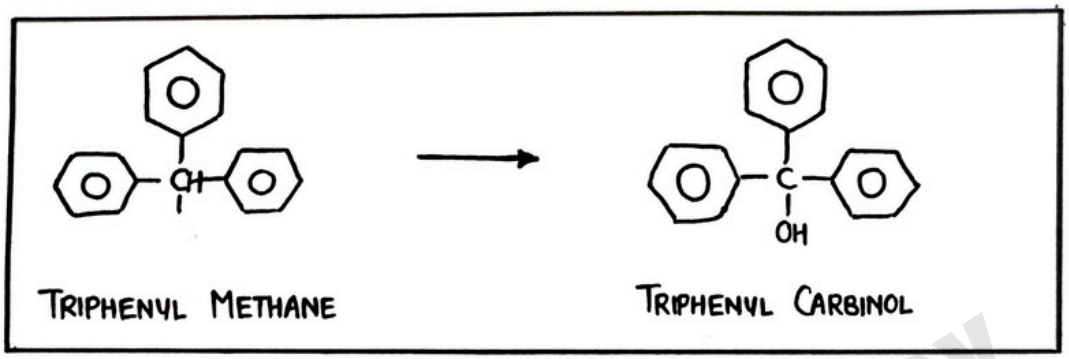


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③ OXIDATION

Triphenyl Methane undergoes oxidation to form Triphenylcarbinol.



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USES OF TRIPHENYL METHANE

- It is used for synthesis of various dyes.
- They also possess Antibacterial & Antifungal properties.
- They are also used as Disinfectants.
- It is also used as Antimicrobial Agent.

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THANK YOU

FOR CHOOSING IMPERFECT PHARMACY AS YOUR STUDY PARTNER

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